World in 2045 Blueprint: Comprehensive Implementation Report

1. Project Goal and Analytical Approach

The primary objective of this project is to answer **“What the world looks like in 2045”** through a strictly **data-driven** methodology 1. The approach involves:

- **Analyzing historical data** dating back to the post–World War 2 era 1.
- **Extrapolating identified trends** to predict future global development 1, 2.
- **Developing a detailed snapshot** of the world in 2045 1.
- **Proposing strategic actions** to encourage positive growth and mitigate negative trajectories 1, 3.

2. Official Scope of Analysis

The analysis covers eleven core dimensions of global development 1, 4:

1. **Population Dynamics:** Country-level growth and demographic metrics 1, 3.
2. **Governance:** Social, political, and governmental development 1, 3.
3. **Quality of Life:** Standard of living and well-being 1, 3.
4. **Health Outcomes:** Mortality and general health trends 1, 3.
5. **Macroeconomics:** Cost of living and economic stability 1, 3.
6. **Technology:** Growth and innovation across various sectors 1, 3.
7. **Inequality:** Income and wealth distribution 1, 3.
8. **Environment:** Climate and environmental risk factors 4, 5.
9. **Geopolitics:** Conflict and stability 4, 5.
10. **Human Capital:** Education and skill levels 4, 5.
11. **Labor:** Market structure and productivity 4, 5.

3. Data Source Inventory

All analysis and predictions are **empirically grounded** in historical data from internationally recognized organizations 4, 5:

- **Demographics:** UN World Population Prospects (WPP), World Bank World Development Indicators (WDI) 6, 7.
- **Governance:** V-Dem, Polity Project, Worldwide Governance Indicators (WGI) 6, 7.
- **Quality of Life:** UNDP Human Development Index (HDI), World Happiness Report (Gallup) 6, 7.
- **Health:** WHO Global Health Observatory (GHO), IHME Global Burden of Disease (GBD) 6, 7.
- **Economics:** IMF World Economic Outlook (WEO), OECD Data, World Bank WDI 6-8.
- **Technology/Innovation:** ITU ICT statistics, WIPO patent data 6, 7.
- **Inequality:** World Inequality Database (WID) 6, 7.
- **Climate/Environment:** UNFCCC, NASA GISS, Climate Risk Indices (Germanwatch) 6, 7.
- **Security:** Uppsala Conflict Data Program (UCDP), SIPRI 9, 10.
- **Education:** UNESCO Institute for Statistics, OECD PISA 9, 10.
- **Labor:** ILOSTAT 9, 10.

4. Target Data Architecture (Google Cloud Platform)

The platform utilizes a **Modern ELT Lakehouse-style warehouse** architecture 10, 11.

- **Cloud Storage (GCS):** Acts as the raw, immutable landing zone for "bronze" files 9, 10.
- **BigQuery:** Serves as the central warehouse, following a **single-dataset convention** for all relations (CI: world2045_ci; dev/prod via environment variables) 9, 10, 12, 13.
- **Orchestration/Ingestion:** Managed via Cloud Run, Cloud Functions, and Cloud Scheduler 9, 10.
- **Transformation:** dbt (Core or Cloud) is used for SQL-based modeling and testing 9, 10.
- **CI/CD:** GitHub Actions for automated building and testing of dbt models 9, 10, 14.

5. ELT Workflow and Layering

The project prioritizes **ELT (Extract, Load, Transform)** to preserve raw data for auditability and reduce pipeline fragility 12, 13.

- **Bronze Layer (Raw):** Immutable tables with source-aligned schemas and minimal typing 15-17. Includes metadata such as ingested_at and source_file 15, 17.
- **Silver Layer (Normalized):** Conformed tables mapped to a **canonical country-year grain** (country_iso3 x year) 16, 18, 19. This layer includes a **fact_country_year_spine** to ensure consistency across domains 8, 20.
- **Gold Layer (Marts):** Integrated, model-ready datasets for feature engineering and forecasting 16, 19.

6. Implementation Phases and Progress

- **Phase 0 (Foundations):** Establishment of the repository, dbt setup, and core dimensions like dim_country (ISO-3 standardized) and dim_year (1945–2100) 20-22.
- **Phase 1 (Backbone):** Ingestion of UN WPP and World Bank WDI. **As of March 2026, the UN WPP 2024 pipeline is stabilized**, with population coverage confirmed for 1950–2100 21-23.
- **Phase 2 (Governance):** Integration of V-Dem and WGI indicators 21, 22.
- **Phases 3–5 (Expansion):** Planned integration of inequality, technology, climate, conflict, and education datasets 21, 22, 24, 25.

7. Conformance and Domain Schemas

Every silver fact table must be **unique on the (country_iso3, year)** primary key 18, 19.

- **Time:** Calendar years are stored as INT64. Multi-year surveys are mapped to a reference year 26, 27.
- **Units:** Explicitly named (e.g., *_pct, *_per_100k). Currency is normalized to *_constant_usd or *_ppp_constant 26, 27.
- **Provenance:** Every silver fact includes source_system, row_hash, and transformation timestamps 28, 29.

Key Domain Indicators include:

- **Population:** Urbanization rate, fertility rate, dependency ratio 28, 29.
- **Health:** Life expectancy, maternal mortality, health expenditure % GDP 28, 29.
- **Economy:** GDP growth, inflation (CPI), government debt % GDP 28, 29.
- **Governance:** V-Dem Liberal Democracy Index, Rule of Law 28, 29.

- **Tech/Education:** Internet usage, R&D % GDP, PISA scores, literacy rates 28, 29.

8. Testing and Modeling Strategy

A robust **dbt test suite** ensures data integrity 30, 31:

- **Structural Tests:** PK uniqueness, not-null constraints, and referential integrity 30, 31.
- **Domain Tests:** Logical checks (e.g., life expectancy < 120) and non-negative constraints for population and GDP 30, 31.
- **Temporal/Coverage Tests:** Outlier detection for extreme year-over-year jumps and missingness heatmaps 32, 33.

Predictive Modeling will utilize:

- Panel regression with fixed effects and Bayesian scenario modeling 32, 33.
- Structural time series for trend/shock decomposition 32, 33.
- Monte Carlo uncertainty bands for long-range geopolitical and development forecasting 32, 33.